

The Parent-Term Wall and the Cost of a Drawn Coordinate

Two structural facts decide what a Rydberg spectrum and an elemental table can be indexed on: an open-shell ionic core carries a separate Rydberg series on each of its levels, all converging on different limits, so a label naming only species and orbital angular momentum can fail to name a channel, and in one compilation of 596 channels it does fail once; and the elemental index closes with defect zero on (row, Z) for any contiguous row partition, the periods and the $n + \ell$ rows alike, while every drawn table — eighteen-column, thirty-two-column or left-step — admits its whole box on its drawn coordinates and has defect box minus cells: 36, 106 and 138.

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Abstract

An index is a set of cells on ordered coordinates. From the cells alone a reader may recover, for each ordered pair of coordinates, how far one reaches given the other; the cells those recovered bounds admit form a closure $\mathcal{R}(X)$, and the *external definition cost* $E(X) = |\mathcal{R}(X)| - |X|$ counts the cells a reader would reconstruct and the index denies. This paper computes E for the elemental index under six presentations and proves why each comes out as it does. On (period, group) the eighteen-column table holds 90 cells in a box of 126 and $E = 36$; the thirty-six are period 1 groups 2–17 and periods 2 and 3 groups 3–12, and the value is forced, because the table occupies both extreme corners of its box and a lemma proved here shows that any two-coordinate index doing so admits its whole box. Of the thirty-six, 25 are slots for an orbital that does not exist ($\ell > n - 1$: 1d, 1p, 2d) and 11 are slots for one that exists and is filled later (3d, and the slot helium vacates); moving helium to group 2 takes E to 20, so sixteen of the thirty-six are the cost of one drawing convention. The same corner lemma gives the thirty-two-column table $E = 106$ and Janet’s left-step table, on its own drawn row and column coordinates, $E = 138$. Against atomic number the picture inverts: with Z as the second coordinate, any partition of Z into contiguous rows laid out in order is a chain in the product order, and a chain is a fixed point of $\mathcal{R}$, so the $n + \ell$ rows give $E = 0$ and the periods give $E = 0$ alike (118 cells in boxes of 944 and 826). The closure therefore distinguishes a coordinate, not a table: it is the group or column coordinate that costs, and it is Z that closes. Both closure theorems are proved and then machine-checked by an SMT solver over every subset of thirteen named boxes. The second half measures the parent-term wall on a compilation of 596 Rydberg channels across 70 species and 28 elements. The LS terms of ℓ^k are counted exhaustively over all 17,476 Slater determinants of the relevant configurations and checked against an independent count; a closed-shell or one-electron core carries one term, a p^2 or p^4 core three, a d^4 core sixteen. 225 rows have a core with more than one level, 132 name the parent in the label and 93 do not, and every one of the 93 belongs to a species printing a single limit. A parent is written in four conventions, and a fifth class of rows writes none; a census run with the commonest convention alone sees 80 of the 139 rows that carry one. Four species print two or more limits, and in each the limits and the parents are in bijection. One outer label, Ba III’s $nd\ 2[3/2]^*\ J=2$, is printed at both of that species’ limits, 17,550.000 cm^{-1} apart, so a label without its parent is two channels there and not a duplicated one.

§0 · The result

Two facts, one about an index and one about a spectrum, and each is a statement about coordinates rather than about chemistry.

The elemental index. Take the 118 elements as cells and ask what their own coordinates imply. Nothing outside the cells enters: for each ordered pair of coordinates (i, j) read off the reach bound $\hat{\varphi}_{ij}(v) = \max\{x_i : x \in X, x_j \leq v\}$, and admit every cell of the observed product that respects all of them. The result is a closure operator $\mathcal{R}$ (Lemma 1, Lemma 2, PROVED), and $E(X) = |\mathcal{R}(X)| - |X|$ is the number of cells the coordinates admit and the index denies — cells whose absence a reader can learn only from outside the index.

presentation	coordinates	cells	box	E
eighteen-column table, f-block set aside	(period, group)	90	126	36
thirty-two-column table	(period, group)	118	224	106
left-step table as drawn	(row $n + \ell$, column)	118	256	138
left-step table as drawn, $Z = 119$ and 120 admitted	(row $n + \ell$, column)	120	256	136
the $n + \ell$ rows against atomic number	$(n + \ell, Z)$	118	944	0
the $n + \ell$ rows, $Z = 119$ and 120 admitted	$(n + \ell, Z)$	120	960	0
the periods against atomic number	(period, Z)	118	826	0

The three drawn tables are forced by Lemma 3: an index on two coordinates that holds both extreme corners of its box admits every cell of that box. The eighteen-column table holds (period 1, group 18) — helium — and (period 7, group 1), so $\mathcal{R}$ is all $7 \times 18 = 126$ cells and E is $126 - 90$ (Theorem 3, PROVED and EXHAUSTIVE). The thirty-two-column layout holds (1, 32) and (7, 1) and gives $224 - 118 = 106$ the same way, and the left-step table on its drawn coordinates — eight rows of $n + \ell$, thirty-two columns with the s-block at the right and helium above beryllium — holds (1, 32) and (8, 1) and gives $256 - 118 = 138$. The three closed lines are forced by Theorem 1: any partition of the atomic numbers into contiguous rows laid out in increasing order is a chain in the product order on (row, Z) , and a chain is a fixed point of $\mathcal{R}$ (Corollary 1). The $n + \ell$ rows are such a partition and so are the periods. The $n + \ell$ index is closed a second, independent way as well: its cells are cut out by two isotone bounds, $Z \leq b(r)$ and $r \leq p(Z)$, and such a set is a fixed point of $\mathcal{R}$ (Theorem 2). Both theorems are proved in §2 and machine-checked over every subset of thirteen finite boxes.

What the closure distinguishes is therefore a coordinate and not an ordering. Under $\mathcal{R}$ nothing separates Janet’s $n + \ell$ rows from Mendeleev’s periods: both close against Z , and both drawings fail on their drawn coordinates by exactly the corner lemma. What does separate them is measured in §4 and is small: the $n + \ell$ row of an element is the largest $n + \ell$ occupied in its tabulated ground configuration at all 108 elements checked, while the period is the largest n occupied at 107 of the 108, palladium being the exception.

The thirty-six are identified one by one (Theorem 4, EXHAUSTIVE). Reading a cell of the eighteen-column layout as the subshell its slot would hold — groups 1–2 the s slot, 3–12 the d slot, 13–18 the p slot — gives

25 cells for an orbital that does not exist, $\ell > n - 1$: **1d** 10, **1p** 5, **2d** 10; 11 cells for an orbital that exists and is occupied later: **3d** 10, and **1s** at period 1 group 2, the slot helium vacates.

Drawing helium at group 2 instead of 18 destroys the first row of gaps and takes E from 36 to 20 (Proposition 1). E is therefore not a property of the elements: sixteen of the thirty-six are the price of one placement decision.

The spectrum. A Rydberg series converges on the ionisation limit of one *level* of the ionic core. A closed-shell core has one term and one level and gives one series per orbital angular momentum ℓ . An open-shell core has several levels, each is an ionisation limit, and the outer electron builds series on each of them — within one parent level and one ℓ , one series for each allowed pair coupling (K, J) — so a label of the form “species + ℓ + outer term”

does not by itself name a channel. When the species prints a single limit at that ℓ and term, the label does determine the channel (Proposition 3); when it prints two, the label may or may not, and in this compilation there is one label at which it does not (a REFUTATION of “the outer label determines the parent”, by the witness Ba III).

Measured on a compilation of 596 channel rows across 70 species and 28 elements:

what the core carries	rows	label names a parent	it does not
bare nucleus	38	0	38
closed shell	159	5	154
one term, one level	174	2	172
one term, two levels	127	77	50
several terms	98	55	43
total	596	139	457

371 rows have a core with a single level and need no parent; 225 have a core with more than one, of which 132 name the parent and 93 do not. **Every one of the 93 belongs to one of thirteen species that prints exactly one limit.** That is the compilation’s own labelling convention — a parent is written on every row of a species that prints two limits — verified to hold on all 596 rows (Proposition 4, EXHAUSTIVE), and it is a fact about how this compilation was labelled, not a theorem about spectra. Four species print two or more limits a wavenumber or more apart (Ba III, Ne I, Ne II, Si I, 99 rows between them), and in all four every row names its parent, one parent per limit, distinct parents at distinct limits.

The census depends on the pattern it is taken with. A parent may be written in four conventions, and a fifth class of rows writes none; they are counted separately (Table 2): a dotted core configuration with the parent level parenthesised (80 rows), the same run together without dots (9), the parent term alone in parentheses (20), and a pair (J_1, j) in which the first entry is the core’s J (30); 457 rows write no parent. A census that recognises only the first convention reports 80 rows of the 139 that carry one, an undercount of 59.

Where the two halves meet. The rows of the $(n + \ell, Z)$ index open at $Z = 1, 3, 5, 13, 21, 39, 57, 89$. The three openings at 21, 57 and 89 are the atomic numbers near which the 3d, 4f and 5f orbitals are described as contracting into the core, and on the 128 measured d and f cells of the quantum-defect index whose core holds no orbital of the cell’s ℓ the mean defect separates across them: median 0.6202 at or past the opening against 0.0335 below it, the collapsed value larger in 773 of the 847 cross pairs, exact two-sided permutation $p = 3.4 \times 10^{-5}$ (Proposition 2, MEASURED). The threshold is read from the index and not fitted. The split controls for ℓ and for the core’s orbital count and not for ionisation stage: all seven collapsed cells are ions, and 63 of the 121 below are neutral or singly ionised.

What is not established.

- **E = 36 is not tested against an independently stated constraint.** The reach bounds on (period, group) are read off the table’s own cells, so the reconstruction and the object it reconstructs have one source. The closure of $(n + \ell, Z)$ is different in kind: it is certified by two structural properties of the cell set, each sufficient on its own, and either can be checked without computing $\mathcal{R}$ at all — but the first of them, chainhood, holds for any contiguous row partition of Z whatever, and so certifies nothing particular to the $n + \ell$ ordering.
- **Why the $n + \ell$ ordering governs filling is not explained here.** The row lengths 2, 2, 8, 8, 18, 18, 32, 32 are $2k'^2$ with $k' = \lceil r/2 \rceil$ for the row $r = n + \ell$, which is arithmetic once the pairing of rows is granted — and that pairing is the whole of what is unexplained; that ground configurations fill in that order is taken from the tabulated configurations and used, not derived. The question is open (Löwdin 1969; Allen and Knight 2003).
- **The collapse class has seven members, all ions.** They are Sc III, Ti III at two multiplicities, Ti IV, Fe VIII, Fe XV and Fe XVI, all at $\ell = 2$ and at spectrum numbers 3 to 16, against 121 below the boundary.

The transition is rapid and not sharp: the largest defect of all 128 is Ca I nd at 0.9084, which sits one below the 3d opening, and Ba II nf is 0.7559 one below the 4f opening. No f cell in the sample lies at or past its opening, and no same-charge comparison across an opening exists in the sample.

- **The absence of an ambiguous row is a property of this compilation’s labelling.** Each of the thirteen species carrying unnamed multi-level cores prints one limit here; that is a fact about which series were captured, not a guarantee that those species have one limit.
- **The counts of rows naming a parent are a floor in the other direction too.** A row may be unambiguous for reasons the printed columns do not carry, and the census cannot see them.
- **The wall’s strongest consequence is not measured here.** That a spectrum with a many-termed core yields published levels and no extractable defect is the structure of parentage (Condon and Shortley 1935; Racah 1943; Cowan 1981) applied to the term counts of §6; what this paper measures is the term counts themselves, the compilation’s own ceiling of spectrum number IV for a multi-level core, and the obstacle counts the index records against d and f cores.

Status words. Seven are used and never merged.

status	meaning
PROVED	a proof is written out below and every step is justified
MACHINE-CHECKED	an SMT solver returned unsat on the negation of an obligation whose variables range over every subset of a named finite box, with both guards passed
EXHAUSTIVE	a decision procedure visited every case in a stated finite family, and the family’s size is printed
SAMPLED	a seeded pseudorandom sweep of a stated size; never exhaustive
MEASURED	a number computed from the stated data by the stated procedure; the paper’s own result, with its sample stated; never a proof
CITED	taken from the literature or from a public database, with the source
REFUTATION	a claim disproved by an explicit witness

§1 • Definitions

D1 (index). Let $d \geq 2$ and let each coordinate $i = 1, \dots, d$ take values in a finite totally ordered set. An *index* is a finite non-empty set X of cells; a cell x has one coordinate value x_i in each of the d sets. Cells are compared coordinatewise: $x \leq y$ means $x_i \leq y_i$ for every i . This is the product order.

D2 (value sets, box). $\hat{A}_i(X) = \{x_i : x \in X\}$ is the set of values coordinate i actually takes. The *box* of X is the product $\prod_i \hat{A}_i(X)$, and $|\text{box}| = \prod_i |\hat{A}_i(X)|$.

D3 (reach bounds). For $i \neq j$ and $v \in \hat{A}_j(X)$,

$$\phi_{ij}(v) = \max \{ x_i : x \in X, x_j \leq v \}.$$

The set is non-empty for every $v \in \hat{A}_j(X)$, since some cell realises the value v itself; so ϕ_{ij} is defined on all of $\hat{A}_j(X)$, and it is non-decreasing in v because enlarging v enlarges the set maximised over.

D4 (the reconstruction $\mathcal{R}$, and the external definition cost E).

$$\mathcal{R}(X) = \{ x \in \prod_i \hat{A}_i(X) : x_i \leq \phi_{ij}(x_j) \text{ for all } i \neq j \}, E(X) = |\mathcal{R}(X)| - |X|.$$

$\mathcal{R}(X)$ is what a reader can reconstruct from the cells alone: the values that occur, and how far each coordinate reaches given each other. $E(X)$ counts the cells that reconstruction admits and X denies. **$E(X) = 0$ is the statement that the index defines itself** — that no fact outside the cells is needed to say which cells there are.

D5 (chain). X is a *chain* when every two of its cells are comparable in the product order.

D6 (Rydberg channel, quantum defect, series limit). A *Rydberg channel* is a sequence of bound levels of one atomic species, one orbital angular momentum ℓ of the outer electron, and one *parent* — one level of the ionic core — with principal quantum number n running and term energies converging on the ionisation limit I of that parent. Write z for the *spectrum number* of the species, the charge the outer electron sees: $z = 1$ for a neutral atom, 2 for a singly ionised one, so that the core has $Z - z$ electrons. With R the Rydberg constant for the species and E_n the level energy, the effective quantum number and the quantum defect are $n^* = z \sqrt{R / (I - E_n)}$ and $\delta_n = n - n^*$ (Rydberg 1890; Ritz 1903; Seaton 1983). A channel's *mean defect* is the average of δ_n over its fitted members, and it is the mean defect that the compilations of §5 and §7 tabulate per channel. Different parents have different I , so **two channels of one species and one ℓ built on two parents converge on two limits.**

D7 (LS term, parent term, level, pair coupling). In LS coupling a configuration is decomposed into terms ^{2s+1}L , and a term into levels by the total angular momentum J , which runs over $|L - S| \dots L + S$. A *parent term* of a channel is the term of the ionic core on which the channel is built; a *parent* in the sense of D6 is one level of that term, with core angular momentum J_1 . For a Rydberg electron of angular momentum ℓ built on a parent level J_1 , the pair coupling of Racah (1942) first couples J_1 with ℓ to K and then K with the electron spin to J ; a series label of the form $nd \ 2[3/2]^\circ \ J=2$ names ℓ , K and J and is one series for each (K, J) on that parent level. The number of terms of a configuration is the number of (L, S) pairs in the decomposition of its multiplicity table, a pair counted as often as it occurs (Theorem 6).

D8 (core class). For a species, the *core* is the ion with one electron removed, in its tabulated ground configuration. A core is:

bare — no electrons remain; *closed* — every occupied subshell is full, so the core is 1S_0 , one term and one level; *one term, one level* — open, but with a single term and that term with a single J ; *one term, two levels* — open, with a single term whose L and S both exceed zero, giving more than one J ; *several terms* — open, with more than one term.

The first three classes have exactly one core level. The last two have more, and only for those can a label that omits the parent fail to name a channel.

D9 (p , the core's orbital count at ℓ). For a channel of orbital angular momentum ℓ , p is the number of occupied subshells of that same ℓ in the core's tabulated ground configuration. $p = 0$ says the core holds no orbital of the channel's ℓ at all, so the outer orbital has no filled counterpart of its own symmetry.

§2 · The reconstruction, and three closure theorems

Lemma 1 (extensivity). $X \subseteq \mathcal{R}(X)$; hence $E(X) \geq 0$.

Proof. Let $x \in X$. Then $x_i \in \hat{A}_i(X)$ for each i , so x lies in the box. Fix $i \neq j$. The cell x itself satisfies $x_j \leq x_i$, so x is one of the cells maximised over in D3, and therefore $\phi_{ij}(x_j) \geq x_i$. All the defining inequalities hold, so $x \in \mathcal{R}(X)$. ■

Lemma 2 (idempotence and monotonicity). $\mathcal{R}(\mathcal{R}(X)) = \mathcal{R}(X)$, and $X \subseteq X'$ implies $\mathcal{R}(X) \subseteq \mathcal{R}(X')$. $\mathcal{R}$ is a closure operator.

Proof. Write $Y = \mathcal{R}(X)$. By Lemma 1, $X \subseteq Y$, and by D4, $Y \subseteq \prod_i \hat{A}_i(X)$; hence $\hat{A}_i(Y) = \hat{A}_i(X)$ for every i , and the two sets have the same box. Let ϕ_{ij} and ψ_{ij} be the reach bounds of X and of Y . Since $X \subseteq Y$, every cell maximised over for ϕ is maximised over for ψ , so $\psi_{ij}(v) \geq \phi_{ij}(v)$. For the reverse, let $y \in Y$ attain $\psi_{ij}(v)$, so $y_j \leq v$. Because $y \in \mathcal{R}(X)$, $y_i \leq \phi_{ij}(y_j)$, and ϕ_{ij} is non-decreasing, so $y_i \leq \phi_{ij}(v)$. Hence $\psi_{ij}(v) \leq \phi_{ij}(v)$, and the two bound families are equal. The defining condition of $\mathcal{R}(Y)$ is then literally the defining condition of $\mathcal{R}(X)$, and $\mathcal{R}(Y) = \mathcal{R}(X) = Y$.

For monotonicity let $X \subseteq X'$ and let $x \in \mathcal{R}(X)$. Then $\hat{A}_i(X) \subseteq \hat{A}_i(X')$ for every i , so x lies in the box of X' ; and for each $i \neq j$ the reach bound of X' maximises over a superset of the cells the bound of X maximises over, so $\phi'_{ij}(x_j) \geq \phi_{ij}(x_j) \geq x_i$. Hence $x \in \mathcal{R}(X')$. $\mathcal{R}$ is extensive, monotone and idempotent: a closure operator in the sense of Moore (1910). ■

Lemma 2 has a consequence worth stating: $E(\mathcal{R}(X)) = 0$ for any X whatever, so an index that is itself a closure — one of the form $\mathcal{R}(Y)$ for some Y — never has a defect. A closure defect is a measurement only when the cells measured are the object and not its closure.

Theorem 1 (a chain is fixed). If X is a chain, then $\mathcal{R}(X) = X$ and $E(X) = 0$.

Proof. Enumerate the chain in order, $x^{(1)} < x^{(2)} < \dots < x^{(N)}$. Each coordinate map $t \mapsto x^{(t)}_i$ is non-decreasing, since $x^{(t)} \leq x^{(t+1)}$ holds coordinatewise.

Let $x \in \mathcal{R}(X)$. For each coordinate i the value x_i lies in $\hat{A}_i(X)$, so the set $\{t : x^{(t)}_i = x_i\}$ is non-empty; let t_i be its largest element. Because coordinate i is non-decreasing in t , $\{t : x^{(t)}_i \leq x_i\}$ is exactly $\{1, \dots, t_i\}$, and therefore

$$\phi_{ij}(x_j) = \max\{x^{(t)}_i : t \leq t_j\} = x^{(t_j)}_i,$$

again by monotonicity of coordinate i . The defining condition of $\mathcal{R}$ thus reads $x_i \leq x^{(t_j)}_i$ for all $i \neq j$.

Let m be an index at which t_j is least, so $t_m \leq t_i$ for every i . Fix any $i \neq m$. On the one hand the condition at (i, m) gives $x_i \leq x^{(t_m)}_i$. On the other hand $x_i = x^{(t_i)}_i$ with $t_i \geq t_m$, and coordinate i is non-decreasing, so $x_i \geq x^{(t_m)}_i$. Hence $x_i = x^{(t_m)}_i$ for every $i \neq m$, and $x_m = x^{(t_m)}_m$ by the choice of t_m . So $x = x^{(t_m)} \in X$. With Lemma 1 this gives $\mathcal{R}(X) = X$. The argument used $d \geq 2$ only to have, for each $i \neq m$, a pair (i, m) available. ■

Corollary 1 (contiguous rows against a totally ordered coordinate). Let Z run over a finite totally ordered set, partition it into blocks each of which is an interval, and number the blocks 1, 2, ... in the order in which they occur. Then the index $X = \{(r(Z), Z)\}$, with $r(Z)$ the number of the block containing Z , satisfies $\mathcal{R}(X) = X$ and $E(X) = 0$.

Proof. If $Z < Z'$ then $r(Z) \leq r(Z')$: the block of Z' occurs no earlier than the block of Z , since the blocks are intervals laid out in order. So $(r(Z), Z) \leq (r(Z'), Z')$ coordinatewise, every two cells are comparable, X is a chain, and Theorem 1 applies. ■

The corollary is the content of the closed lines of §0's table. It holds for the $n + \ell$ rows, for the periods, and for any other way of cutting the atomic numbers into consecutive runs; it says nothing about which cut is the right one.

Theorem 2 (bi-monotone bounds are fixed). Let $d = 2$, let A and B be finite totally ordered sets, let $F : B \rightarrow A$ and $G : A \rightarrow B$ be non-decreasing, and let

$$X = \{(a, b) \in A \times B : a \leq F(b) \text{ and } b \leq G(a)\}.$$

Then $\mathcal{R}(X) = X$, and $E(X) = 0$.

Proof. By Lemma 1 it suffices to show $\mathcal{R}(X) \subseteq X$. Let $(a, b) \in \mathcal{R}(X)$. Then $a \leq \phi_{12}(b)$ and $b \leq \phi_{21}(a)$, and $(a, b) \in \hat{A}_1(X) \times \hat{A}_2(X) \subseteq A \times B$.

Let $y = (y_1, y_2) \in X$ attain $\hat{\varphi}_{12}(b)$, so $y_2 \leq b$ and $y_1 = \hat{\varphi}_{12}(b)$. Since $y \in X$, $y_1 \leq F(y_2)$, and F is non-decreasing with $y_2 \leq b$, so $y_1 \leq F(b)$. Hence $a \leq \hat{\varphi}_{12}(b) \leq F(b)$. The same argument with the coordinates exchanged gives $b \leq \hat{\varphi}_{21}(a) \leq G(a)$. So (a, b) satisfies both defining inequalities of X and lies in $A \times B$; therefore $(a, b) \in X$. ■

Remark (the converse, for $d = 2$). Every closed two-coordinate index is of Theorem 2's form: if $\mathcal{R}(X) = X$, then $X = \{(a, b) \in \hat{A}_1(X) \times \hat{A}_2(X) : a \leq \hat{\varphi}_{12}(b), b \leq \hat{\varphi}_{21}(a)\}$ by D4, and $\hat{\varphi}_{12}$ and $\hat{\varphi}_{21}$ are non-decreasing by D3, so $F = \hat{\varphi}_{12}$ and $G = \hat{\varphi}_{21}$ serve. So at $d = 2$ the structures $\mathcal{R}$ recovers are exactly the sets cut out by a pair of isotone bounds, one per ordered pair of coordinates, and an index whose true shape is such a pair loses nothing by being handed to a reader as a bare set of cells.

Lemma 3 (the corners fill the box). Let $d = 2$ and write $a-, a+$ for the least and greatest elements of $\hat{A}_1(X)$ and $b-, b+$ for those of $\hat{A}_2(X)$. If $(a-, b+) \in X$ and $(a+, b-) \in X$, then $\mathcal{R}(X) = \hat{A}_1(X) \times \hat{A}_2(X)$: the whole box.

Proof. Let (a, b) be any cell of the box. For the first condition: $(a+, b-) \in X$ and $b- \leq b$, so $a+$ is among the values maximised over in $\hat{\varphi}_{12}(b)$, whence $\hat{\varphi}_{12}(b) = a+ \geq a$. For the second: $(a-, b+) \in X$ and $a- \leq a$, so $b+$ is among the values maximised over in $\hat{\varphi}_{21}(a)$, whence $\hat{\varphi}_{21}(a) = b+ \geq b$. Both conditions hold, so $(a, b) \in \mathcal{R}(X)$. With $\mathcal{R}(X) \subseteq \text{box}$ by definition, the two sets are equal. ■

The machine check. Theorems 1 and 2 and Lemma 3 are decidable over a fixed finite box: quantify over every subset X of the box that realises every value of both coordinates — so that the box is its own $\hat{A}(X)$ by D2 and the maxima of D3 are attained — encode membership in $\mathcal{R}$ by the witness form

└ $x \in \mathcal{R}(X)$ if and only if, for every ordered pair $i \neq j$, there is a cell $y \in X$ with $y_j \leq x_j$ and $y_i \geq x_i$,

which is equivalent to D3 without naming a maximum, and ask a solver to refute the negation of the implication. The encoding is guarded twice before any obligation is reported. The **fidelity guard** is SAMPLED: on 300 pseudorandom subsets, seed 7, drawn from five box shapes, three implementations of membership in $\mathcal{R}(X)$ decide every cell of the observed box — the solver's own term, evaluated with the drawn subset substituted for its Boolean variables; a staircase operator implemented independently of this paper; and a third implementation written directly from D3 and D4 — 3,578 cell decisions, 0 disagreements between any two of the three; the same comparison applied to a deliberately wrong encoding, with strict inequality $y_j < x_j$ in place of $y_j \leq x_j$, disagrees on 2,007 of the 3,578, so the guard is not vacuous. The **non-vacuity guard** is run once for each of the thirteen (hypothesis, box) pairs below, the 7×18 box included: each hypothesis is satisfiable by a subset that is strictly smaller than its box and holds at least two cells, so that unsat on the negation is not unsat on an empty hypothesis. Thirteen obligations were then discharged, all returning unsat:

obligation	boxes
Theorem 1: chain $\Rightarrow \mathcal{R}$ -closed	$3 \times 3, 4 \times 4, 5 \times 5, 6 \times 6, 3 \times 3 \times 3, 2 \times 2 \times 2 \times 2$
Theorem 2: bi-monotone $\Rightarrow \mathcal{R}$ -closed (F, G unknown isotone tables)	$3 \times 3, 4 \times 4, 5 \times 5$
Lemma 3: both corners $\Rightarrow$ whole box	$3 \times 3, 4 \times 4, 5 \times 5, 7 \times 18$

Status: **MACHINE-CHECKED** over the boxes named, in addition to PROVED in general. The 7×18 box of Lemma 3 is the box of the eighteen-column elemental index itself, so the corner argument of §3 is machine-checked on its own object over every subset that realises every period and every group.

§3 • The elemental index on (period, group)

Take the eighteen-column layout (Mendeleev 1869, in its modern form) with the f-block set aside: period 1 holds groups 1 and 18; periods 2 and 3 hold groups 1, 2 and 13–18; periods 4 through 7 hold all eighteen. That is $2 + 8 + 8 + 4 \times 18 = 90$ cells.

Theorem 3. For the eighteen-column layout X , $\mathcal{R}(X)$ is the full 7×18 box and $E(X) = 126 - 90 = 36$.

Proof. $\hat{A}_1(X) = \{1, \dots, 7\}$ and $\hat{A}_2(X) = \{1, \dots, 18\}$, since period 4 alone realises every group and group 1 alone realises every period. The extreme corners are (period 1, group 18), which is helium, and (period 7, group 1), which is francium; both are cells of X . Lemma 3 applies and gives $\mathcal{R}(X) = \{1, \dots, 7\} \times \{1, \dots, 18\}$, of size 126. Hence $E = 126 - 90 = 36$. ■

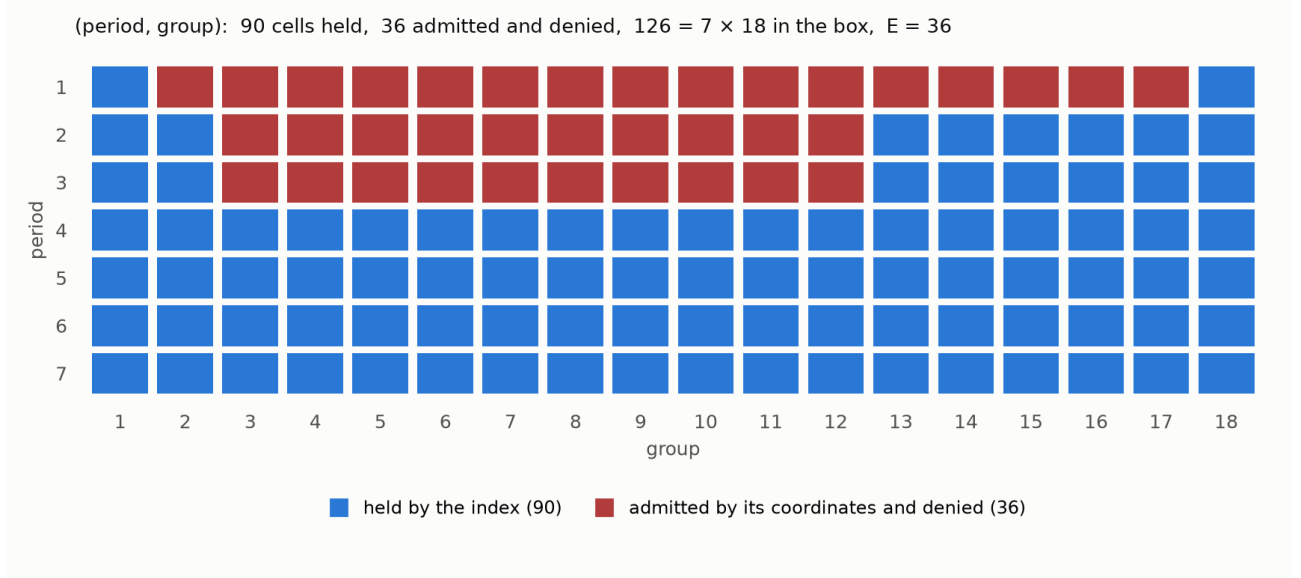


Figure 1. The eighteen-column presentation as an index. Blue: the ninety cells it holds. Red: the thirty-six its own coordinates admit and it denies. A reader given only the occupied cells reconstructs all one hundred and twenty-six.

The same computation run directly, as a staircase closure over the ambient product, returns $|\mathcal{R}(X)| = 126$ and $E = 36$, and a second, independently written implementation of D3–D4 returns the same set; both are EXHAUSTIVE over the 126-cell box. The two reach bounds are constant — $\phi(\text{group} \mid \text{period} \leq p) = 18$ for every p and $\phi(\text{period} \mid \text{group} \leq g) = 7$ for every g — which is Lemma 3’s conclusion read as a pair of bounds.

Theorem 4 (the thirty-six, identified). The thirty-six cells of $\mathcal{R}(X) \setminus X$ are period 1 groups 2–17 (16 cells), period 2 groups 3–12 (10) and period 3 groups 3–12 (10). Reading a cell (p, g) as the subshell its slot would hold — $\ell = 0$ for $g \leq 2$, $\ell = 2$ for $3 \leq g \leq 12$, $\ell = 1$ for $g \geq 13$, and $n = p$ — they divide as

the thirty-six	count
forbidden by $\ell \leq n - 1$: 1d 10, 1p 5, 2d 10	25
an orbital exists, and is occupied later: 3d 10, and 1s at (1, 2)	11

Proof. The three lists are read off $\mathcal{R}(X) \setminus X$, which Theorem 3 computes; enumerating the 36 cells and applying the slot rule and the test $\ell > n - 1$ is a finite check over 36 cases, all visited. For the two boundary readings: at period 1 the groups 3–12 give $\ell = 2$ with $n = 1$, and $2 > 0$, so all ten are forbidden; groups 13–17 give $\ell = 1$ with $n = 1$, and $1 > 0$, so those five are forbidden, and there are five rather than six because group 18 at period 1 is occupied by helium and is not a gap. The cell (1, 2) gives $\ell = 0$ with $n = 1$, and $0 \leq 0$, so it is not forbidden: it is the 1s slot, and it is empty only because helium is drawn at group 18. At period 2 the ten d slots give $\ell = 2$ with $n = 2$ and $2 > 1$, forbidden; at period 3 they give $\ell = 2$ with $n = 3$ and $2 \leq 2$, not forbidden — the 3d subshell exists and is filled at period 4 rather than period 3. Hence $10 + 5 + 10 = 25$ forbidden and $10 + 1 = 11$ deferred, and $25 + 11 = 36$. ■

Status: EXHAUSTIVE over the 36 cells.

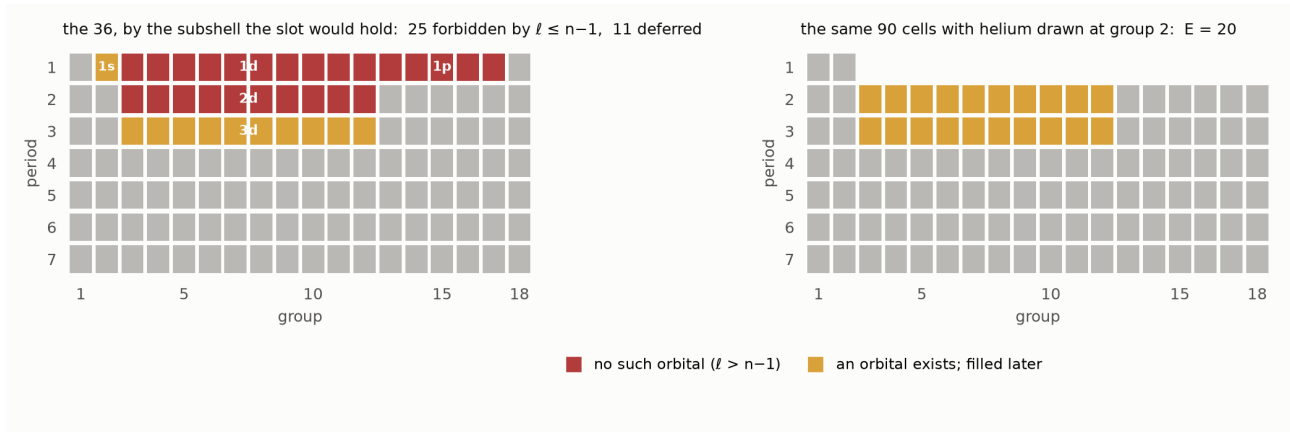


Figure 2. Left: the thirty-six coloured by the subshell each slot would hold — 25 red where no orbital of that ℓ exists at that n , 11 gold where one exists and is occupied later. Right: the same ninety cells with helium drawn at group 2. The entire first row of gaps disappears and E falls to 20, the twenty being periods 2 and 3, groups 3–12.

Proposition 1 (helium’s placement costs sixteen cells). Drawing helium at group 2 rather than 18, with all other cells unchanged, leaves 90 cells and gives $E = 20$; the twenty admitted cells are periods 2 and 3, groups 3–12.

Proof. With helium at group 2 the period-1 cells are (1, 1) and (1, 2), so $\phi(\text{group} \mid \text{period} \leq 1) = 2$ rather than 18. The other reach bound is unchanged: $\phi(\text{period} \mid \text{group} \leq g) = 7$ for every g , since group 1 is occupied at every period. The reconstruction is therefore $\{(p, g) : g \leq \phi(\text{group} \mid \text{period} \leq p)\} = \{(1, 1), (1, 2)\} \cup (\{2, \dots, 7\} \times \{1, \dots, 18\})$, of size $2 + 108 = 110$, and $E = 110 - 90 = 20$. The twenty cells of $\mathcal{R}(X) \setminus X$ are the ten d slots at each of periods 2 and 3, since periods 4–7 are full and period 1 now contributes nothing. Lemma 3 does not apply, because the corner (period 1, group 18) is no longer occupied. ■

The difference, $36 - 20 = 16$, is what it costs a reader to be told where helium is drawn. Helium’s position is the most argued question in periodic-table design: the eighteen-column convention places it above neon, and the left-step form and a quantum-chemical case place it above beryllium (Scerri 2020). E measures the definitional cost of the choice and returns no verdict on it.

Two further presentations, for scale. The thirty-two-column layout, with the f-block inserted and all 118 elements drawn, has $\hat{A}_1 = \{1, \dots, 7\}$, $\hat{A}_2 = \{1, \dots, 32\}$, holds the corners (1, 32) and (7, 1), and so by Lemma 3 admits its whole 224-cell box: $E = 224 - 118 = 106$. Widening the drawing raises the defect. And adjoining a third coordinate, the block, to the eighteen-column layout — a coordinate that is a function of the group and is not monotone in it — raises E from 36 to 100. A coordinate carrying information the index already has is not free.

§4 • The same elements against atomic number, and the left-step table

Janet’s left-step table (Janet 1928, 1929; Stewart 2010) draws the elements in rows of constant $n + \ell$, with the blocks in the order f, d, p, s from left to right, the s-block at the right-hand edge and helium above beryllium; it is the drawing of the $n + \ell$ filling rule that Madelung (1936) published later and Klechkovskii (1962) justified. Order the subshells by $n + \ell$, and within one value of $n + \ell$ by n . Filling them in that order, with capacity $2(2\ell + 1)$ each, partitions the atomic numbers $1 \dots 118$ into eight contiguous blocks, one per value of $n + \ell$; write $r(Z)$ for the value of $n + \ell$ of the block containing Z . The rows have lengths

| 2, 2, 8, 8, 18, 18, 32, 30

— the last truncated only because the drawing stops at 118; admitting $Z = 119$ and 120 gives 32 and the sequence 2, 2, 8, 8, 18, 18, 32, 32, which is $2k'^2$ for $k' = 1, 1, 2, 2, 3, 3, 4, 4$ (EXHAUSTIVE: the eight products are checked) — that is, $k' = \lceil r/2 \rceil$, and the pairing of consecutive rows into equal lengths is where the content lies. The rows open at

$$\mid Z = 1, 3, 5, 13, 21, 39, 57, 89$$

and end at $Z = 2, 4, 12, 20, 38, 56, 88, 118$.

This section closes two different indices built from that drawing and keeps them apart. The first is the drawing itself, on its drawn coordinates (row, column). The second replaces the column by the atomic number: its cells are the pairs $(r(Z), Z)$.

The drawing on its drawn coordinates. Number the columns 1 ... 32 with the f-block at 1–14, the d-block at 15–24, the p-block at 25–30 and the s-block at 31–32. Row 1 holds hydrogen at column 31 and helium at column 32; row 8 opens with actinium at column 1. So the index holds the corners (1, 32) and (8, 1) of its 8×32 box, Lemma 3 applies, and $\mathcal{R}(X)$ is the whole box: $E = 256 - 118 = 138$, or $256 - 120 = 136$ when $Z = 119$ and 120 are admitted (EXHAUSTIVE over the 256-cell box, both implementations). On its drawn coordinates the left-step table costs a reader more than the eighteen-column table does, for the same reason and with a larger box.

Theorem 5. The 118 cells of the $(n + \ell, Z)$ presentation satisfy $\mathcal{R}(X) = X$ and $E(X) = 0$, and this follows in two independent ways.

Proof. *First route.* The cells form a chain. If $Z < Z'$ then $r(Z) \leq r(Z')$, because r is non-decreasing by construction — a row is a contiguous block of atomic numbers and the rows are laid out in increasing order — so $(r(Z), Z) \leq (r(Z'), Z')$ coordinatewise. Every two cells are therefore comparable, and Theorem 1 gives $\mathcal{R}(X) = X$; this is Corollary 1 with the $n + \ell$ rows as the blocks. Chainhood was verified over all $C(118, 2) = 6,903$ pairs (EXHAUSTIVE).

Second route. The cells are cut out by two isotone bounds. Put $b(r) = \max\{Z : (r, Z) \in X\}$, so $b = (2, 4, 12, 20, 38, 56, 88, 118)$, and $p(Z) = \max\{r : (r, Z') \in X \text{ for some } Z' \leq Z\} = r(Z)$. Both are non-decreasing, and

$$\mid X = \{ (r, Z) : 1 \leq r \leq 8, 1 \leq Z \leq 118, Z \leq b(r), r \leq p(Z) \}$$

— verified cell by cell over the 8×118 grid (EXHAUSTIVE). Theorem 2 then gives $\mathcal{R}(X) = X$ directly. ■

Both routes were also run numerically: the staircase closure and an independent implementation of it each return $|\mathcal{R}(X)| = 118$ and $E = 0$, and the same holds for the 120-cell version in a box of 960.

The periods close against Z in exactly the same way. The eighteen-column table's periods are a contiguous partition of the same 118 atomic numbers into rows of 2, 8, 8, 18, 18, 32, 32, opening at $Z = 1, 3, 11, 19, 37, 55, 87$. By Corollary 1 the index $\{(\text{period}(Z), Z)\}$ is a chain and is closed: 118 cells in a box of $7 \times 118 = 826$, $E = 0$ (EXHAUSTIVE, both implementations). Corollary 1 was also run on every one of the 117 two-row contiguous partitions of $Z = 1 \dots 118$; all 117 are closed under both implementations. **The closure does not prefer the $n + \ell$ rows to the periods.** What it registers is the second coordinate: against Z any monotone row coordinate closes, and against a group or column coordinate every drawn table is a corner-filled box.

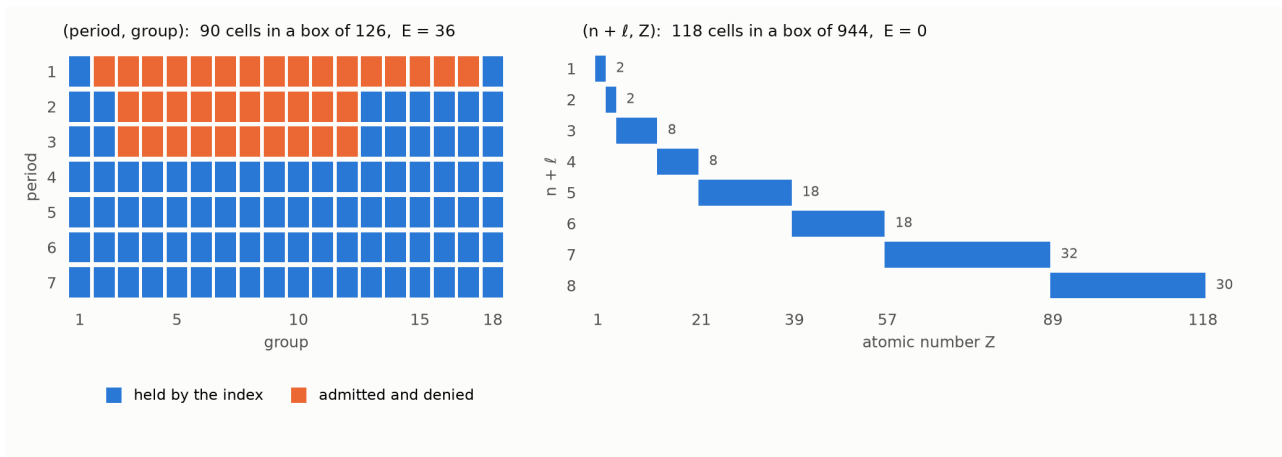


Figure 3. The same elements on two presentations. Left: $(\text{period}, \text{group})$, 90 cells in a box of 126, $E = 36$, with the thirty-six admitted cells in orange. Right: $(n + \ell, Z)$, 118 cells in a box of 944, $E = 0$, drawn as its eight rows with their lengths. The boxes are over unlike coordinates and their sizes are not comparable; what the right-hand panel shows is a chain, and the periods drawn against Z would show one too.

Closure follows the shape, not the count. Appending a further cell that preserves the chain — for instance (3, 13), a hypothetical thirteenth member of the third row — leaves 119 cells and $E = 0$. Moving one element to the wrong row instead — deleting (5, 30) and inserting (2, 30) — breaks the chain and takes E to 54. E is a measurement of arrangement.

What does separate the two row coordinates is how each is read off a configuration. For every Z from 1 to 108 the largest value of $n + \ell$ over the occupied subshells of the tabulated ground configuration equals the $n + \ell$ row of Z : 108 agreements, 0 exceptions (EXHAUSTIVE over the 108 configurations). The largest value of n over the occupied subshells equals the period at 107 of the 108, the exception being palladium, $Z = 46$, whose tabulated ground configuration $4d^{10}$ occupies no $n = 5$ subshell and which sits in period 5 (EXHAUSTIVE). Read instead from the *differentiating* electron alone — the one subshell whose occupancy rises from $Z - 1$ to Z — the $n + \ell$ identification fails at exactly six elements, $Z = 25, 30, 43, 47, 48$ and 80 , each the successor of an s-to-d rearrangement in which the added electron lands in the s subshell that the previous element had emptied. The row is a property of the whole configuration and not of the electron most recently added.

Chainhood is sufficient, not necessary. The subshells themselves, indexed by $(n + \ell, \ell)$ for $n \leq 7$ and $\ell \leq 3$, give 22 cells in a box of 40 with $E = 0$; its recovered bounds are $\ell \leq \min(\lfloor (n + \ell - 1)/2 \rfloor, 3)$ and $(n + \ell) \leq 7 + \ell$, both isotone, so Theorem 2 applies. It is not a chain: $(n + \ell, \ell) = (4, 1)$ and $(5, 0)$ are incomparable. Theorem 2 covers cases Theorem 1 does not.

§5 · Where the rows open, and what the spectra do there

The three highest row openings of the $(n + \ell, Z)$ index are $Z = 21, 57$ and 89 . The literature on orbital collapse — the contraction of a 3d, 4f or 5f orbital from a diffuse outer orbital into a compact inner one as the nuclear charge rises (Fermi 1928; Goeppert-Mayer 1941; Griffin, Andrew and Cowan 1969; Connerade 1978) — places the collapse near those atomic numbers and makes its position depend on the ionisation stage; it does not give integers. The three integer thresholds used below are the index's own row openings, and they are therefore **read off the index and not fitted to the spectra**; that is the whole of their claim to independence.

They are close to, and not identical with, the first occupation of the orbital in a tabulated ground configuration. 3d is first occupied at $Z = 21$, which is the opening exactly; 4f at $Z = 58$, one past the opening at 57, because lanthanum at 57 takes 5d; and 5f at $Z = 91$, two past the opening at 89, because actinium and thorium take 6d (EXHAUSTIVE over $Z = 2 \dots 108$, first occupation recorded for every subshell).

The sample. The quantum-defect index of §7's second data set holds 148 measured cells at $\ell = 2$ or 3, a cell being keyed by $(Z, \text{spectrum number}, 2S + 1, \ell)$ and carrying the mean defect of the channel fitted there (D6); Ti III nd at two multiplicities is two cells. Of the 148, 128 have $p = 0$ — the core holds no orbital of the cell's ℓ (D9). Seven of the 128 sit at or past the opening of the corresponding row (21 for $\ell = 2$, 57 for $\ell = 3$) and 121 below it.

Proposition 2 (the mean defect separates across the openings). On the 128 cells the medians are 0.6202 at or past the opening and 0.0335 below it. The collapsed value exceeds the uncollapsed one in 773 of the $7 \times 121 = 847$ cross pairs, with no tied cross pair, so the Mann–Whitney statistic is $U = \min(773, 74) = 74$. The exact two-sided permutation p , from the distribution of the rank sum over all $C(128, 7) = 94,525,795,200$ assignments of seven cells to the upper class with midranks for the 12 tied values (Mann and Whitney 1947), is 3.4×10^{-5} (one-sided 1.7×10^{-5}). The tie-corrected normal approximation without continuity correction, reported for comparison, gives $z = -3.66$ and $p = 2.5 \times 10^{-4}$; the sign of z is the min-U convention and carries no direction.

Status: **MEASURED**, sample 128 cells, split fixed in advance by the index and not fitted. The seven are Sc III nd, Ti III nd at two multiplicities, Ti IV nd, Fe VIII nd, Fe XV nd and Fe XVI nd, with mean defects 0.6533, 0.7899, 0.7987, 0.6202, 0.4000, 0.1278 and 0.0750.

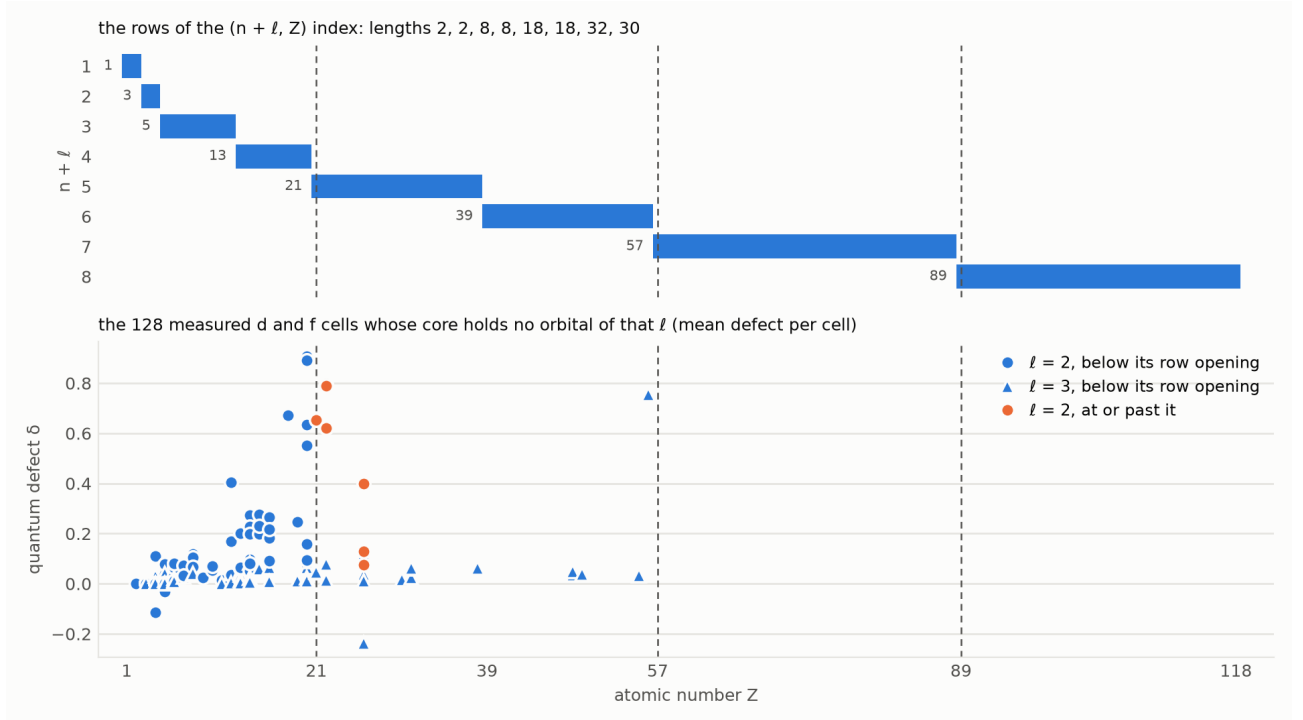


Figure 4. Upper: the eight rows of the $(n + \ell, Z)$ index, with the three highest openings marked at $Z = 21$, 57 and 89. Lower: the 128 measured d and f cells whose core holds no orbital of the cell's ℓ , mean defect against Z . Orange marks the seven at or past the opening of their row.

Three cautions the data themselves impose. First, the transition is rapid and not sharp: the single largest defect among the 128 is Ca I nd at 0.9084, at $Z = 20$, one below the 3d opening, and Ba II nf is 0.7559 at $Z = 56$, one below the 4f opening. Second, the split controls for ℓ and for p and not for ionisation stage: every one of the seven collapsed cells is an ion, at spectrum numbers 3, 3, 3, 4, 8, 15 and 16, while of the 121 below, 34 are neutral and 29 singly ionised — 63 at spectrum number 2 or less — and ionisation stage is what the collapse literature identifies as moving the collapse to lower Z , so the two classes differ in charge as well as in position. Two of the seven,

Fe XV nd at 0.1278 and Fe XVI nd at 0.0750, carry smaller defects than Ca I. The one isoelectronic sequence in the sample that crosses an opening is the argon-core nd series: K I 0.2460 at $Z = 19$, Ca II 0.6341 at $Z = 20$, Sc III 0.6533 at $Z = 21$ — the large step is from 19 to 20, one below the opening, and the step across the opening itself is small. Third, the pair the contrast is usually stated on, Ti IV nd at 0.6202 against Sr II nf at 0.0618, a factor of 10.0, both with $p = 0$, is a pair differing in Z , in spectrum number and in ℓ ; it illustrates the size of the effect and controls for nothing.

The same structure appears in the index’s own account of which of its 104,832 cells could be measured. The index records, against each unmeasured cell, a named obstacle rather than a probability, and three of the obstacles are parent counts: *open-shell core*, 3 parents on 9,756 cells, 16 parents on 11,605, 119 parents on 5,280 — 26,641 cells in all. The index was built with each cell’s core configuration taken in Madelung ($n + \ell$) order, and with the core so taken, recomputing its configuration at each of those cells and counting its LS terms (§6) shows the three numbers to be exactly the maxima of the three blocks: every cell bounded at 3 has a core with one open p subshell and 3 terms; every cell bounded at 16 has one open d subshell, with 5, 8 or 16 terms; every cell bounded at 119 has one open f subshell, with 7, 17, 47, 73 or 119 (EXHAUSTIVE over the 26,641 cells). With the core taken instead in its tabulated ground configuration, as D8 defines it, the identification holds on 21,271 of the 26,641: 4,675 cells have a core with two open subshells in the tabulated configuration, 565 a closed core, and 130 a d core under the f bound. **The obstacle to measuring a quantum defect, stated cell by cell, is a term count** — of the configuration the index’s builder assigned, and the tabulated configurations agree with it on four cells in five.

§6 · The parent-term wall

The rule. A Rydberg series is defined against one ionisation limit, and an ionisation limit is the energy of one level of the ionic core (D6). A core with k levels therefore offers k limits, and the outer electron at a given ℓ builds series on each of them; within one parent level and one ℓ there are several series, one for each allowed pair (K, J) in Racah’s pair coupling (D7), and all of them converge on that level’s limit. Series built on different parent levels converge on different limits. The series interleave in energy and are distinguished not by ℓ but by the parent (Condon and Shortley 1935; Racah 1942, 1943; Cowan 1981). The compilation of §7 shows the rule in its own rows: Ba III’s core $^2P^\circ$ has two levels, 20 rows converge on the lower limit and 2 on the upper, and the 20 are several series at one limit.

The number of parent levels is a term count and a level count, and both are finite and computable.

Theorem 6 (the terms of ℓ^k). Enumerating every Slater determinant of ℓ^k — every k -subset of the $2(2\ell + 1)$ spin-orbitals — and peeling the resulting multiplicity table from its highest entry gives the LS terms exactly. Over $\ell = 0, 1, 2, 3$ and all k from 0 to $4\ell + 2$, this is 17,476 determinants, and the term counts are these.

Table 1 — the number of LS terms of ℓ^k .

ℓ^k	$k = 0$	1	2	3	4	5	6	7	8	9	10	$k = 11$... 14
p^k	1	1	3	3	3	1	1					
d^k	1	1	5	8	16	16	16	8	5	1	1	
f^k	1	1	7	17	47	73	119	119	119	73	47	17, 7, 1, 1

Proof. Write m_ℓ and m_s for the one-electron projections and $M_\ell = \sum m_\ell$, $M_s = \sum m_s$ for their totals. By the exclusion principle (Pauli 1925) the determinants of ℓ^k are in bijection with the k -subsets of the $2(2\ell + 1)$ spin-orbitals (m_ℓ, m_s), and every such determinant is an eigenvector of M_ℓ and M_s . The multiplicity table $N(M_\ell, M_s)$ counting them is therefore exact. The determinant space decomposes into LS multiplets, and a multiplet ^{2s+1}L contributes

exactly one state to each (M_l, M_s) with $|M_l| \leq L$ and $|M_s| \leq S$, both running in unit steps, and none outside that rectangle (CITED: the representation theory of angular momentum, Condon and Shortley 1935, ch. III and VII). So N is a sum of rectangles, one per term, each with its corner at (L, S) .

The highest-weight step. Let N be a non-zero sum of such rectangles, and let (μ_0, ν_0) be the positive entry of largest M_s and, among those, largest M_l . Some rectangle with corner (L, S) contains (μ_0, ν_0) , so $S \geq \nu_0$ and $L \geq \mu_0$. If $S > \nu_0$, that rectangle also contains (μ_0, S) , which is then a positive entry with larger M_s , contradicting the choice; so $S = \nu_0$. If then $L > \mu_0$, the rectangle contains (L, ν_0) , a positive entry at the same M_s with larger M_l , again a contradiction; so $L = \mu_0$. Hence a term with $L = \mu_0$ and $S = \nu_0$ is present, and subtracting its rectangle leaves a sum of rectangles with one term fewer, every entry still non-negative. Induction on the number of terms shows that the peel — take the entry of largest M_s then largest M_l that is still positive, record a term with $L = M_l$ and $S = M_s$, subtract its rectangle — removes exactly one term per pass, never drives an entry negative, and terminates with the table identically zero, having recorded the configuration's multiset of terms. The run confirms both consequences: 17,476 decrements over every ℓ and k , none below zero, and an independent count of the same multiset by second differences, the number of terms with given (L, S) being $N(L, S) - N(L+1, S) - N(L, S+1) + N(L+1, S+1)$, agrees for every ℓ^k (EXHAUSTIVE). ■

Status: **PROVED**, given the CITED decomposition into multiplets, and **EXHAUSTIVE** over 17,476 determinants by two independent decision procedures. The p and d rows agree with the tables of Condon and Shortley (1935) and the f row with the tables of Nielson and Koster (1963) (CITED); the counts are symmetric under $k \mapsto 4\ell + 2 - k$, particle-hole conjugation, at every entry (EXHAUSTIVE).

Two corollaries carry the wall.

Corollary 2. A closed subshell contributes one term, 1S , and a configuration all of whose subshells are closed has exactly one term and one level. A single electron or a single hole in one subshell also gives one term — checked for $\ell = 0, 1, 2, 3$ (EXHAUSTIVE) — with two levels when $\ell > 0$. **A closed-shell core therefore gives exactly one Rydberg series per ℓ and per outer term, and a label of the form “species + ℓ + term” names it unambiguously.**

Corollary 3. A p^2 or p^4 core carries 3 terms and 5 levels, a d^4 core 16 terms, an f^6 core 119; a p^5 or d^1 core one term and two levels (EXHAUSTIVE for the level counts named). Such a core offers that many limits, and its Rydberg spectrum decomposes into interleaved series, each converging on the limit of the parent level it is built on.

Proposition 3 (a single limit determines the channel). Let a series label name species, orbital angular momentum and the outer term or pair-coupling labels, but no parent. If the species' captured spectrum prints a single limit at that ℓ and label, the channel the label denotes is determined: the parent is that limit's level, and the label plus the species supplies every datum of D6.

Proof. The parent of a channel is a level of the core, and each level has its own limit (D6). If only one limit is printed at that ℓ and label, every captured member with that label converges on it, so the parent is the level of that limit, and the channel is the sequence of those members. ■

The converse does not hold in general, and this paper does not claim it. When two limits are present, the outer label may still determine the parent: in pair coupling, a K value that couples to one core level only fixes J_1 , and a label carrying such a K or J is admissible on one parent alone. What holds is weaker and is a fact about this data:

REFUTATION (the outer label does not determine the parent). Stripping the parent prefix from every one of the 596 labels — the three prefix conventions of Table 2; the (J_1, j) pair carries its parent's J_1 inside the pair and is left intact — and keying on (species, stripped label), exactly two keys carry more than one limit, and only one of the two is a parent case: Ba III's $nd\ 2[3/2]^* J=2$ is printed at 289,100.000 and at 306,650.000 cm^{-1} (EX-

HAUSTIVE over 596 rows; the other key is Ca II's, whose two limits differ by 0.010 cm^{-1} and are one limit at two roundings). The label admits two distinct channels, with distinct limits and hence distinct defects, and does not determine one. The claim “the outer label always determines the parent” is refuted by that witness.

Proposition 3 is a statement about the data, not about physics: a species may have a multi-level core whose second limit simply was not captured. The census in §7 measures which case each row of the compilation is in.

§7 • The census

The data. A compilation of measured Rydberg channels: 596 rows under an eleven-column header, 119 of them two-member channels, across 70 species and 28 elements, 3,342 levels and 2,269 interior cells. Each row carries the species, the series label, the range of n , the level and interior-cell counts, a containment column, the range of effective quantum number, the mean defect and its spread, the spectrum number and the series limit in cm^{-1} . Every count below is recomputed from the table itself; the row count was checked against an independent reader of the same table (EXHAUSTIVE, 596 rows, row indices identical).

Provenance. Every level in the compilation is fitted to the NIST Atomic Spectra Database (Kramida, Ralchenko, Reader and the NIST ASD Team 2024). The compilation's own source list attributes 25 of the 70 species to a named compilation — Al I and Al II to Kaufman and Martin (1991), Be I to Kramida and Martin (1997), Na I and K I to Sansonetti (2008a, 2008b), and twenty species to NIST ASD directly — and names Sugar and Corliss (1985) and Sugar and Musgrove (1990, 1995) as additional sources for ionisation limits; the remaining 45 species carry no species-level attribution beyond the database itself (EXHAUSTIVE over the source list). The second data set used in §5, the quantum-defect index of 104,832 cells, records a provenance string on every cell; among its measured cells 25 are NIST ASD retrievals and 5 are the Hartree–Slater asymptotic defects of Theodosiou, Inokuti and Manson (1986) (EXHAUSTIVE).

Table 2 — the four conventions in which a parent is written, and the rows that write none.

convention	example	rows
dotted core configuration, parent level parenthesised	5p5. (2P*<3/2>) . nd 2[3/2]* J=2	80
the same run together, without dots	2s22p5(2P*3/2)nd 2[3/2]* J=1	9
the parent term alone, in parentheses	(³ P)ns ⁴ P J=5/2	20
a (J_1, j) pair, the first entry the core's J	nd (3/2, 5/2)* J=3	30
no parent written	nd 2D J=3/2	457
carrying a parent		139

The label forms are those of the NIST Atomic Spectra Database (Martin and Wiese 1996). The (J_1, j) pair is J_1j coupling notation; its first entry fixes the parent *level* only because every core it is written on has a single term, which is the case for all 30 rows. The $2[K]^\circ$ labels are Racah's pair coupling (D7). Operationally, “the label names a parent” means “the label contains a parenthesised group”, and the four conventions are the four shapes that group takes. A census taken with the first pattern alone — a dotted configuration followed by a parenthesised group — sees 80 of those 139 and reports 59 rows as parentless that are not. **The conventions have to be read off the material before a parent is counted**, and the margin between the narrowest and the widest reading is 59 rows of the 139.

The core census. For each of the 70 species the core is the ion at one fewer electron, in its tabulated ground configuration; its class (D8) is decided by counting the terms of its open subshell with Theorem 6. **No core in the compilation has more than one open subshell** — the six open shapes present are s^1 , p^1 , p^2 , p^4 , p^5 and d^1 (EX-

HAUSTIVE over the 70 species) — so a term count here is a single-subshell count and Theorem 6 gives it exactly, with no recoupling of two open shells to consider. Three ions whose ground configuration is not the configuration the electron count alone would give are taken from their published ground levels: Ti III's core Ti IV is $3d^2D_{3/2}$, Zn II's core Zn III is $3d^{10}^1S_0$, and Hg II's core Hg III is $5d^{10}^1S_0$ (CITED, NIST ASD). The result is the table of §0, split here by whether the row's label names a parent.

Table 3 — the 596 rows by core class and by whether a parent is written.

core	rows	parent named	not named
bare nucleus	38	0	38
closed shell	159	5	154
one term, one level	174	2	172
one term, two levels	127	77	50
several terms	98	55	43

371 rows sit in the first three classes, where the core has one level and no parent needs writing; 7 of those write one anyway — C II's 5 rows, whose closed $2s^2$ core is written ($1S$), and Ga II's 2. The remaining 225 have a core with more than one level, 132 name the parent and 93 do not.

Proposition 4 (the compilation's labelling convention holds on every row). The compilation writes a parent on every row of a species that prints two or more limits a wavenumber or more apart, and on no other row is a parent required. On all 596 rows: every row of the four two-limit species names a parent, and each of the 93 rows whose core has more than one level and whose label names none belongs to one of thirteen species — Al IV, Ar I, Ar II, C I, F I, K II, Mg III, N I, N II, O III, P II, S III, Ti III — each of which prints exactly one limit on every row it has. By Proposition 3 the parent of each of the 93 is determined.

Status: **EXHAUSTIVE**, 596 rows, 93 rows and 13 species, every row's limit compared. The proposition confirms that a labelling rule was applied without exception; it is not a property of the thirteen spectra. Ar II is the concrete case: its core $3p^4$ has 3 terms and 5 levels (Corollary 3), and all 44 of its rows — 30 naming no parent and 14 naming one — print one limit, so the compilation captured Ar II series on one parent level only.

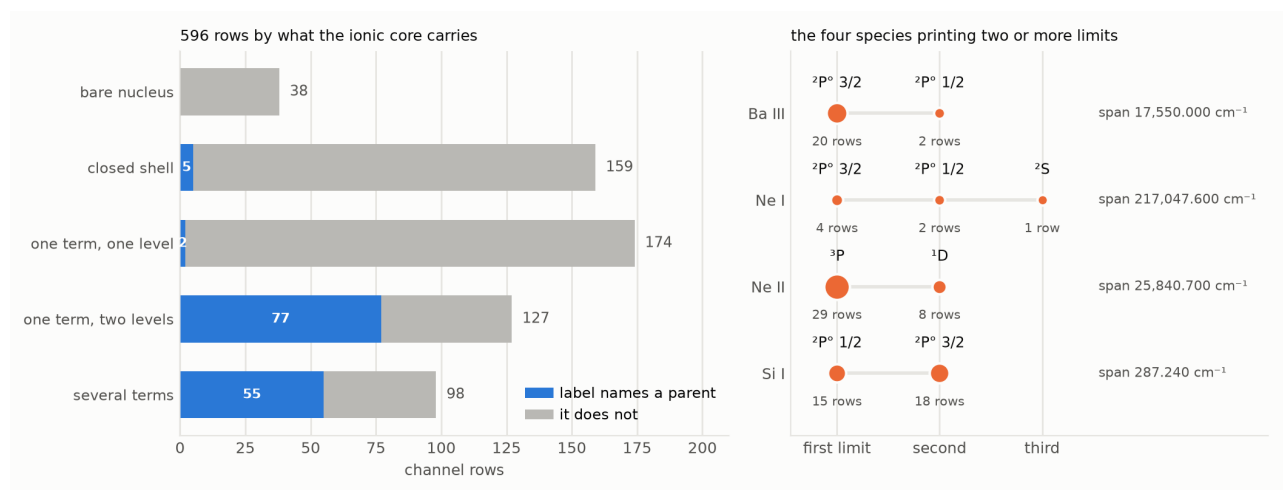


Figure 5. Left: the 596 rows by what the ionic core carries, with the part of each class whose label names a parent in blue. Right: the four species printing two or more limits, with the parent at each limit, the number of rows converging on it, and the span from the lowest limit to the highest in exact arithmetic on the printed values.

The four species with more than one limit. Seven species print more than one value in the limit column. Three of the seven differ by less than a wavenumber — Ca II by 0.010, Li I by 0.036, Zn I by 0.020 cm^{-1} — and are one limit written at two roundings, not two parents; they are excluded on that ground and not on any judgement about the physics. The remaining four are these.

Table 4 — the four species printing two or more limits, and the parent at each.

species	limits (cm^{-1})	separation (cm^{-1})	rows at each	parent
Ba III	289,100.000 / 306,650.000	17,550.000	20 / 2	$2P^{\circ}_{3/2} / 2P^{\circ}_{1/2}$
Ne I	173,929.750 / 174,710.090 / 390,977.350	+780.340 / +217,047.600	4 / 2 / 1	$2P^{\circ}_{3/2} / 2P^{\circ}_{1/2} / 2S$
Ne II	330,388.600 / 356,229.300	25,840.700	29 / 8	$3P / 1D$
Si I	65,747.760 / 66,035.000	287.240	15 / 18	$2P^{\circ}_{1/2} / 2P^{\circ}_{3/2}$

Every printed limit carries three decimals, and every separation is computed in exact rational arithmetic from the printed values and printed to those three decimals. **In all four species every row names its parent, each limit carries exactly one parent, and distinct limits carry distinct parents** (EXHAUSTIVE over the 99 rows). The pattern of the parents is the wall itself in two forms: in Ba III, Ne I and Si I the two lowest limits are the two levels $2P^{\circ}_{3/2}$ and $2P^{\circ}_{1/2}$ of a *single* term of a p^5 core, so the splitting is fine structure; in Ne II the two limits are two *terms*, $3P$ and $1D$, of a p^4 core, which Theorem 6 says has three; and Ne I's third limit, 217,047.600 cm^{-1} above its first, belongs to a different core configuration altogether, $2s2p^6 2S$.

The ceiling. No species in the compilation has a core with more than one level above spectrum number IV, and IV is reached once, by Al IV, whose core is $2p^5 2P^{\circ}$. The two highest-charge species present are Fe XV, whose core is a one-level $3s 2S_{1/2}$, and Fe XVI, whose core is closed. The compilation reaches spectrum number 16 and stops at 4 for multi-level cores; that is a fact about which series were captured, and it is consistent with, though not predicted by, the interleaving of Corollary 3.

And the same ceiling appears in what can be measured at all. Of the 104,832 cells of the quantum-defect index, 61,152 have $Z \leq 92$; of those 11,416 also have spectrum number ≤ 10 and $Z \leq 83$; of those 4,395 also have $\ell \leq 4$ and spectrum number ≤ 6 ; and of those **1,925 also have a core with a single term** in its tabulated ground configuration (D8), or 1,755 with the core taken in Madelung order as the index was built — the 170 cells between the two counts have cores Pd, Ce and Pt, whose tabulated configurations are closed or one-term and whose Madelung-order configurations are open (EXHAUSTIVE over the 104,832 cells). The last cut is the wall, and it removes 2,470 of the 4,395 cells that survive the first three (2,640 in Madelung order). The cuts are stated here as the ones that produce the counts; the simpler cuts $Z \leq 92$, spectrum number ≤ 10 , $\ell \leq 4$ and a one-term tabulated core give 61,152, 12,720, 7,950 and 3,550 (EXHAUSTIVE).

§8 · Repeats that are not repeats

The wall has a practical consequence for anyone auditing such a table: **two rows that look like a duplicate may be two channels.** The clearest instance in this data is Ba III.

Ba III's core is Ba IV, ground configuration ending $5p^5$, a single hole in a p subshell and therefore by Corollary 2 a single term, $2P^{\circ}$, with two levels $J = 3/2$ and $J = 1/2$. The compilation prints Ba III channels at two limits, 289,100.000 and 306,650.000 cm^{-1} , 17,550.000 cm^{-1} apart to the precision at which the limits are printed. The two limits are the two levels of that term by construction — the rows at the lower limit name $2P^{\circ}_{3/2}$ and those at

the upper name ${}^2P_{1/2}$ — so their difference is the compilation’s fine-structure interval for the core and not an independent test of it. The outer label $nd\ 2[3/2]^* J=2$ is printed at both, and it is the only outer label in the table of which that is true for a genuine pair of limits (§6). Under the rule of §6 those two rows are **two series with two defects, not one series entered twice**, and there is nothing to repair.

The genuine repeats in the same table are of three kinds, and each was identified by reading the rows rather than by matching a pattern.

One series printed under two notations. Normalising a label by stripping its parent prefix and folding superscript typography, and keying on (species, series, n-range), the table holds exactly **four** repeated keys, all Ar II. Each pair agrees in n-range, level count, interior count, effective-quantum-number range, mean defect to four decimals, spectrum number and limit, and differs only in notation, in the containment column and in the spread $\sigma(\delta)$. Ar II prints one limit on all 44 of its rows, so no Ar II row is ambiguous about its parent and none of the four is a parent case (EXHAUSTIVE).

One channel fitted in two adjacent windows of n. Keying on (species, series, limit), Ba III holds two repeats, and in each the two rows’ n-ranges are adjacent and disjoint: 5–7 with 8–22, and 6–8 with 9–23. Both halves of each pair carry the same parent. These are one channel fitted twice, at low n and at high n (EXHAUSTIVE).

Two rows nothing printed separates. Keying on (species, series, n-range) over the raw labels, the table holds exactly two repeated keys, both Si I: $nd\ (3/2, 3/2)^* J=1$ at $n = 20\text{--}50$ and $nd\ (3/2, 5/2)^* J=3$ at $n = 20\text{--}56$. Each pair differs in exactly two of the eleven printed columns — the containment column, and the fourth decimal of δ : +0.0126 against +0.0129, and +0.0570 against +0.0573 (EXHAUSTIVE). The key cannot separate them because nothing printed does.

The moral is narrow and worth stating. **A duplicate is a reading, not a pattern match.** The four notation pairs, the two window pairs, the two unseparated Si I pairs and the Ba III parent pair all look alike to a key that ignores the limit and the parent, and they have four different dispositions once the limit and the core’s term structure are read.

§9 · What the wall means for bracketing

The weakest useful statement one can make about a Rydberg series is that a level with a measured neighbour on each side lies between them. That containment is a deduction rather than a fit — it needs no ionisation limit, no quantum defect and no functional form — and it survives everything that leaves the order of the levels intact. A companion paper in this series defines and tests it; here only its precondition is at issue, and the parent-term wall bears on that precondition in one specific way, and the direction is favourable.

The bracket is a statement about *one* series. Its hypothesis is that the three levels are consecutive members of a monotone sequence. Where an open-shell core interleaves several series in one energy region, three levels adjacent in the published list need not be consecutive members of any one series, and the bracket’s hypothesis fails not because the physics is different but because the levels have been sorted by energy across parents. So the wall sets the bracket’s domain: it may be applied within a channel and not across an energy region, and identifying the channel is exactly identifying the parent.

That is why the census of §7 is a precondition and not a curiosity. In this compilation 225 rows have a core with more than one level, and had any of the 93 that write no parent belonged to a species with two limits, the levels behind that row could not be assigned to a series without further information, and no containment test run on them would mean anything. The census finds none, so every row of the compilation is a channel in the sense of D6, and a containment statement has a well-defined object at every one of its 2,269 interior cells. The compilation’s own containment column records the outcome of its test per row as m of k, and its totals are these: 1,577 passing cells

of 1,738 on 392 rows, with 78 rows carrying no three consecutive members and 126 rows whose test conditions the data do not meet (EXHAUSTIVE over the column). The test itself is not defined or examined here; the figures are the column's, reproduced from it. The second of the two refusals is, in part, the wall again: a test that cannot reconstruct the parent-and-coupling selection behind a row declines to run rather than guessing it.

The general form of the point is this. **An index may bracket a quantity only along an axis on which the quantity is monotone**, and a Rydberg term is monotone in n within a parent and not across parents. The parent is therefore not a refinement of the channel label. It is part of the coordinate, and an index that omits it has, by exactly the measure of §2, cells it cannot define.

§10 • Verification record

The machine checks behind this paper discharge 145 obligations, all passing — 117 EXHAUSTIVE, 15 GUARD (one SAMPLED fidelity guard, its negative control, and thirteen non-vacuity guards) and 13 MACHINE-CHECKED — and a self-test adds six negative controls, each of which must be and is refuted. Every number printed in this paper is produced there or is CITED.

object	status	family or box	count
Lemma 1 (extensivity), Lemma 2 (idempotence, monotonicity), Corollary 1, the Remark after Theorem 2	PROVED	—	—
Theorem 1 (chain $\Rightarrow$ closed)	PROVED + MACHINE-CHECKED	every observing subset of 3×3 , 4×4 , 5×5 , 6×6 , $3\times 3\times 3$, $2\times 2\times 2\times 2$	6
Theorem 2 (bi-monotone $\Rightarrow$ closed)	PROVED + MACHINE-CHECKED	every isotone (F, G) on 3×3 , 4×4 , 5×5	3
Lemma 3 (corners $\Rightarrow$ whole box)	PROVED + MACHINE-CHECKED	every observing subset of 3×3 , 4×4 , 5×5 , 7×18	4
Theorem 3, $E(\text{period, group}) = 36$	PROVED + EXHAUSTIVE	the 126-cell box, two independent implementations	—
Theorem 4, the 36 identified	EXHAUSTIVE	36 cells	—
Proposition 1, helium at group 2	PROVED + EXHAUSTIVE	the 126-cell box	—
the thirty-two-column table and the left-step drawing, $E = 106$ and 138	PROVED (Lemma 3) + EXHAUSTIVE	the 224- and 256-cell boxes	—
Theorem 5, $E(n + \ell, Z) = 0$, both routes	PROVED + EXHAUSTIVE	6,903 pairs; the 8×118 grid	—
Corollary 1 on the periods, and on every two-row partition	PROVED + EXHAUSTIVE	the 826-cell box; 117 partitions	—
the two row coordinates read from the ground configurations	EXHAUSTIVE	108 configurations, twice	—
Theorem 6, the terms of ℓ^k	PROVED (multiplet decomposition CITED) + EXHAUSTIVE	17,476 Slater determinants; 17,476 decrements; the second-difference count	—
the channel census (§7), and that no core has two open subshells	EXHAUSTIVE	596 rows, 70 species	—
Proposition 4, the labelling convention on every row	EXHAUSTIVE	596 rows; 93 rows, 13 species	—

object	status	family or box	count
the four two-limit species, limits in exact arithmetic	EXHAUSTIVE	99 rows	—
the repeats of §8, and the one outer label printed at two limits	EXHAUSTIVE	596 rows under four keys	—
the REFUTATION of “the outer label determines the parent”	REFUTATION	the witness Ba III	—
the index’s obstacle counts as term counts (§5), under both configuration rules	EXHAUSTIVE	26,641 cells	—
the measurable-cells chain (§7), under both configuration rules and under the simpler cuts	EXHAUSTIVE	104,832 cells	—
Proposition 2, the collapse	MEASURED	128 cells, split fixed by the index; the exact test over C(128, 7) assignments	—
the fidelity guard	SAMPLED	300 draws, seed 7, five box shapes, 3,578 cell decisions	—
term tables of p^k , d^k (Condon and Shortley 1935) and f^k (Nielson and Koster 1963)	CITED	—	—
ground configurations, ground levels, measured levels	CITED	NIST ASD and the compilations of §7	—

The two guards. No machine-checked obligation is reported unless both pass. *Encoding fidelity* (SAMPLED): the solver’s term for membership in $\mathcal{R}$, the independently implemented staircase operator and a third implementation written from D3–D4 are evaluated on 300 pseudorandom subsets of five box shapes, seed 7 — 3,578 cell decisions, 0 disagreements between any two. The same comparison run against a deliberately altered operator (strict inequality in place of $\leq$) disagrees on 2,007 of the 3,578, so the guard can fail and does when it should. *Non-vacuity*: for each of the thirteen (hypothesis, box) pairs, the chain, corner or bi-monotone hypothesis is shown satisfiable by a subset that is neither the whole box nor smaller than two cells, so a refuted negation is not a refuted emptiness.

The negative controls, each a REFUTATION. “E(period, group) = 35” is refuted, the value being 36. “The index with one element moved to the wrong row is still closed” is refuted, E being 54; the control *moves* a cell rather than appending one, and the distinction is the content of Theorem 1 — appending (3, 13) preserves the chain, so E stays 0 and an appended cell would be no control at all. “ d^4 carries 15 terms” is refuted, the count being 16. “The left-step table is closed on its drawn (row, column) coordinates” is refuted, E being 138. “The exact rank test reaches $p < 0.05$ on two interleaved triples” is refuted, the exact p being 7/10. And “every subset of the 3×3 box that realises every coordinate value is $\mathcal{R}$ -closed” is refuted by the solver, which returns sat with an explicit witness.

What is not machine-checked, and why. The channel census, the term enumeration and the elemental indices are decided by exhaustive enumeration over finite families whose sizes are printed above; an SMT encoding would add nothing, since there is no quantifier over an unknown set. Proposition 2 is a statistic on a sample of 128 and is marked MEASURED, never EXHAUSTIVE: its class sizes are 7 and 121, its split is fixed in advance by the index rather than chosen to maximise the separation, its p is the exact permutation value, and the confound with ionisation stage is printed beside it. The decomposition of a many-electron space into LS multiplets, on which Theorem 6’s proof rests, is cited and not re-proved.

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